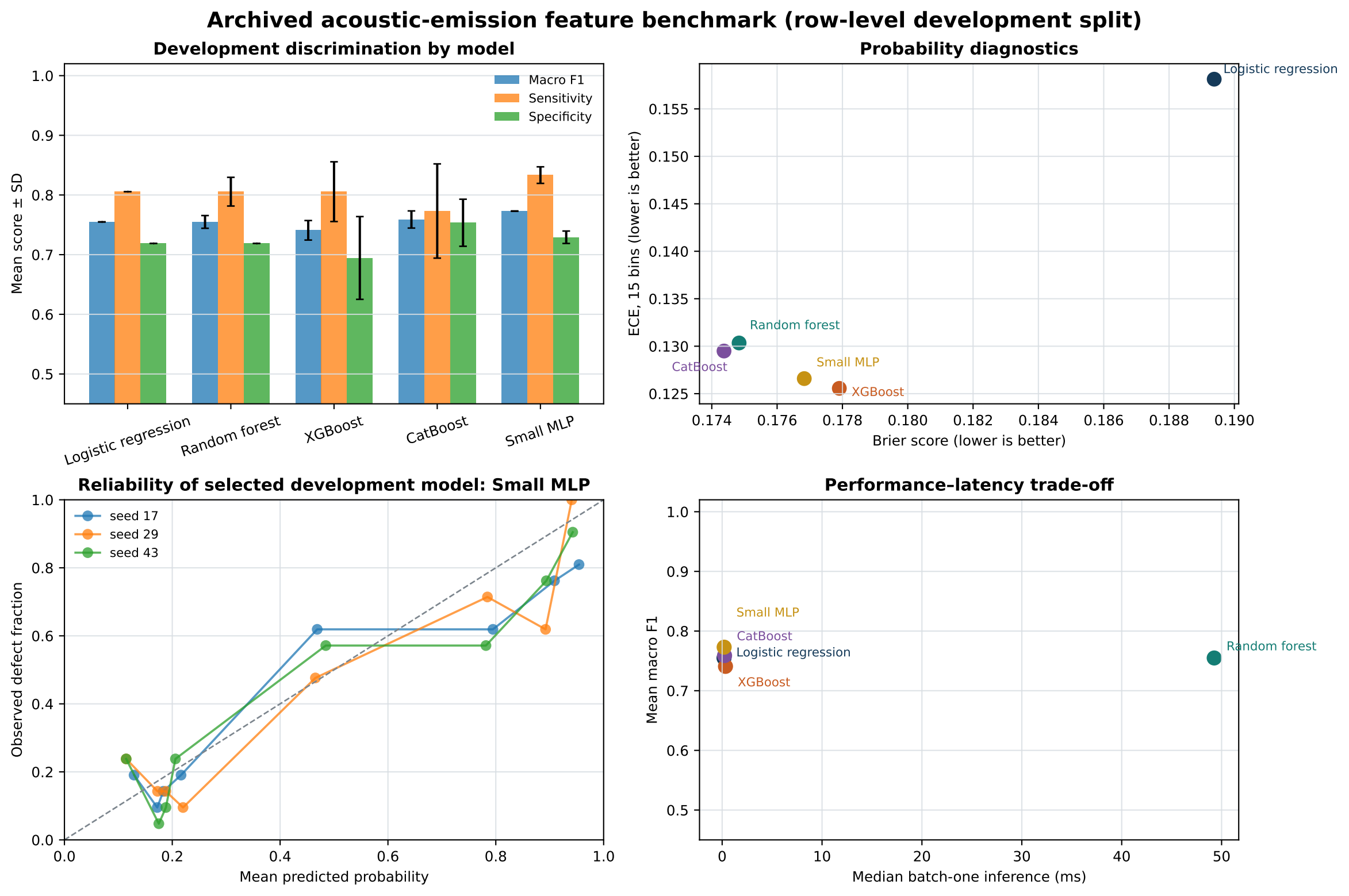
Archived acoustic-emission feature benchmark (row-level development split)
Development discrimination by model
Macro F1
Sensitivity
Specificity
Mean score ± SD
1.0
0.9
0.8
0.7
0.6
0.5
Logistic regression
Random forest
XGBoost
CatBoost
Small MLP
Probability diagnostics
ECE, 15 bins (lower is better)
0.155
0.150
0.145
0.140
0.135
0.130
0.125
Logistic regression
Random forest
Small MLP
CatBoost
XGBoost
0.174
0.176
0.178
0.180
0.182
0.184
0.186
0.188
0.190
Brier score (lower is better)
Reliability of selected development model: Small MLP
seed 17
seed 29
seed 43
Observed defect fraction
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Mean predicted probability
Performance–latency trade-off
Mean macro F1
1.0
0.9
0.8
0.7
0.6
0.5
Small MLP
CatBoost
Logistic regression
XGBoost
Random forest
0
10
20
30
40
50
Median batch-one inference (ms)